

DevOps- Assignment-3

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Configure the given flask application to run on port 6000.
Create a docker image of this application.
Create a repository on docker hub and push the above image to the docker repository.
Deploy the application on minikube.
Create a service for the deployment and check if it displays the application on the service URL.

Now modify the flask application.
Create a version2 docker image of this application.
Push the docker image to the above repository with a different tag.
On minikube set the new image for the deployment so that when you refresh the earlier page and make sure it is showing the new page.

1. Configuring the flask app

```
uadmin@ubuntu1: ~/devops3/flask_mysql_
uadmin@ubuntu1:~$ cd devops-3
bash: cd: devops-3: No such file or directory
uadmin@ubuntu1:~$ cd devop3
bash: cd: devop3: No such file or directory
uadmin@ubuntu1:~$ ls
ai.html  Desktop  devops-s11  Downloads  Public  update-webapp.sh
demo-app  devops3  Dockerfile  Music  snap  Videos
demo.txt  devops-4  Documents  Pictures  Templates
uadmin@ubuntu1:~$ cd devops3
uadmin@ubuntu1:~/devops3$ ls
flask_mysql_app
uadmin@ubuntu1:~/devops3$ cd flask_mysql_app
uadmin@ubuntu1:~/devops3/flask_mysql_app$ ls
app.py  db_init.py  docker-compose.yml  README.md  requirements.txt  templates
uadmin@ubuntu1:~/devops3/flask_mysql_app$ sudo nano app.py
[sudo] password for uadmin:
uadmin@ubuntu1:~/devops3/flask_mysql_app$ sudo nano db_init.py
```

```
GNU nano 6.2 app.py
from flask import Flask, render_template, request, redirect, url_for, session
import mysql.connector

app = Flask(__name__)
app.secret_key = "change_this_secret_key"

DB_CONFIG = {
    "host": "flask-db",
    "user": "appuser",
    "password": "apppassword",
    "database": "userdb"
}
```

```
@app.route("/logout")
def logout():
    session.clear()
    return redirect(url_for("home"))

if __name__ == "__main__":
    app.run(host="0.0.0.0", port=6000, debug=True)
```

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^G Help

^X Exit

^O Write Out

^R Read File

^W Where Is

^_ Replace

^K Cut

^U Paste

^T Execute

^J Justify

2. Creating a docker image of this application.

Dockerfile

```
GNU nano 6.2 Dockerfile
FROM ubuntu
RUN apt update -y
RUN DEBIAN_FRONTEND=noninteractive apt install python3 -y
RUN apt install python3-pip python3-venv -y
RUN python3 -m venv /opt/venv
ENV PATH="/opt/venv/bin:$PATH"
RUN pip3 install flask
RUN pip3 install mysql-connector-python
RUN mkdir /app
COPY app.py /app/
COPY templates /app/templates
CMD ["python3", "/app/app.py"]
```

```
uadmin@ubuntu1: ~/devops3/flask_mysql_app
uadmin@ubuntu1:~/devops3/flask_mysql_app$ docker build -t pritimaurya/devops-assignment3 .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
Install the buildx component to build images with BuildKit:
https://docs.docker.com/go/buildx/

Sending build context to Docker daemon 14.34kB
Step 1/12 : FROM ubuntu
----> 53958ec7b67c
Step 2/12 : RUN apt update -y
----> Using cache
----> 4332bf99b4c9
Step 3/12 : RUN DEBIAN_FRONTEND=noninteractive apt install python3 -y
----> Using cache
----> f0aa1e873e25
Step 4/12 : RUN apt install python3-pip python3-venv -y
----> Using cache
----> 8508944bb250
Step 5/12 : RUN python3 -m venv /opt/venv
----> Using cache
----> 156cdb56cf89
Step 6/12 : ENV PATH="/opt/venv/bin:$PATH"
----> Using cache
----> 412a2d3c72bc
Step 7/12 : RUN pip3 install flask
----> Using cache
----> b4436628a668
Step 8/12 : RUN pip3 install mysql-connector-python
----> Using cache
----> 09d435f420f9
Step 9/12 : RUN mkdir /app
----> Running in 12e4920f0cb9
----> Removed intermediate container 12e4920f0cb9
----> d0370b741e22
Step 10/12 : COPY app.py /app/
----> 1e4e9cdb2515
Step 11/12 : COPY templates /app/templates
----> f86580bbbd7c
Step 12/12 : CMD ["python3", "/app/app.py"]
----> Running in 580d299fa1e3
----> Removed intermediate container 580d299fa1e3
----> 6d8abf459a26
Successfully built 6d8abf459a26
Successfully tagged pritimaurya/devops-assignment3:latest
```

3. Create a repository on docker hub and push the above image to the docker repository.

```
uadmin@ubuntu1: ~/devops3/flask_mysql_app
Successfully built 6d8abf459a26
Successfully tagged pritimaurya/devops-assignment3:latest
uadmin@ubuntu1:~/devops3/flask_mysql_app$ docker push pritimaurya/devops-assignment3
Using default tag: latest
The push refers to repository [docker.io/pritimaurya/devops-assignment3]
3e49383002cf: Mounted from pritimaurya/flask-app
81e2f2053c8f: Mounted from pritimaurya/flask-app
1bd0c9faeb55: Mounted from pritimaurya/flask-app
96729d8d5401: Mounted from pritimaurya/flask-app
f5bcd819baf1: Pushed
d1f56e4c7f2f: Mounted from pritimaurya/flask-app
331d1419ab3e: Mounted from pritimaurya/flask-app
7873277a23e7: Mounted from pritimaurya/flask-app
7ade7b7856ac: Mounted from pritimaurya/flask-app
1f65d64983f1: Pushed
d19c9e709049: Pushed
latest: digest: sha256:6d8abf459a26ebf57db70e63317b797c0b98e6b9ffee3ac5f0441ed7bb4b0e5b size: 2917
uadmin@ubuntu1:~/devops3/flask_mysql_app$ minikube status
minikube
type: Control Plane
host: Stopped
kubelet: Stopped
apiserver: Stopped
kubeconfig: Stopped

uadmin@ubuntu1:~/devops3/flask_mysql_app$ minikube start
minikube v1.38.1 on Ubuntu 22.04 (vbox/amd64)
Using the docker driver based on existing profile
Starting "minikube" primary control-plane node in "minikube" cluster
Pulling base image v0.0.50 ...
Restarting existing docker container for "minikube" ...
Preparing Kubernetes v1.35.1 on Docker 29.2.1 ...
Verifying Kubernetes components...
  Using image gcr.io/k8s-minikube/storage-provisioner:v5
Enabled addons: default-storageclass, storage-provisioner
Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
```

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DevOps assignment 3--- Priti Maurya, PRN: 8033---- and ---- Sai Raut PRN: 8038----

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This repository contains 1 tag(s).

Tag	OS	Type	Pulled	Pushed
latest		Image	less than 1 day	32 minutes

See all

4. Deploy the application on minikube:
 - Creating service for the deployment

```
uadmin@ubuntu1:~/devops3/flask_mysql_app$ kubectl create deployment flask-app --image=pritiimaurya/devops-assignment3:latest --port=6000 --replicas=2
deployment.apps/flask-app created
uadmin@ubuntu1:~/devops3/flask_mysql_app$ kubectl expose deployment devops-assignment3 --type=NodePort
Error from server (NotFound): deployments.apps "devops-assignment3" not found
uadmin@ubuntu1:~/devops3/flask_mysql_app$ kubectl expose deployment pritiimaurya/devops-assignment3 --type=NodePort
error: there is no need to specify a resource type as a separate argument when passing arguments in resource/name form (e.g. 'kubectl get resource/<resource_name>' instead of 'kubectl get resource resource/<resource_name>')
uadmin@ubuntu1:~/devops3/flask_mysql_app$ kubectl expose deployment pritiimaurya/devops-assignment3 --type=NodePort --port=6000
error: there is no need to specify a resource type as a separate argument when passing arguments in resource/name form (e.g. 'kubectl get resource/<resource_name>' instead of 'kubectl get resource resource/<resource_name>')
uadmin@ubuntu1:~/devops3/flask_mysql_app$ kubectl expose deployment flask-app --type=NodePort
service/flask-app exposed
uadmin@ubuntu1:~/devops3/flask_mysql_app$ minikube service flask-app
```

NAMESPACE	NAME	TARGET PORT	URL
default	flask-app	6000	http://192.168.49.2:30441

Opening service default/flask-app in default browser...

```
uadmin@ubuntu1:~/devops3/flask_mysql_app$ Gtk-Message: 12:03:06.718: Not loading module "atk-bridge": The functionality is provided by GTK natively. Please try to not load it.
```



5. Now modifying the flask application,
 - Creating a version2 docker image of this application

```
uadmin@ubuntu1: ~/devops3/flask_mysql_app

uadmin@ubuntu1:~/devops3/flask_mysql_app$ ls
app.py      docker-compose.yml  README.md      templates
db_init.py  Dockerfile          requirements.txt

uadmin@ubuntu1:~/devops3/flask_mysql_app$ cd templates
uadmin@ubuntu1:~/devops3/flask_mysql_app/templates$ ls
dashboard.html  index.html  login.html  register.html

uadmin@ubuntu1:~/devops3/flask_mysql_app/templates$ sudo nano index.html
uadmin@ubuntu1:~/devops3/flask_mysql_app/templates$ docker build -t pritiimaurya/devops-assignment3:v2
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
Install the buildx component to build images with BuildKit:
https://docs.docker.com/go/buildx/

docker: 'docker build' requires 1 argument

Usage:  docker build [OPTIONS] PATH | URL | -

Run 'docker build --help' for more information
uadmin@ubuntu1:~/devops3/flask_mysql_app/templates$ cd ..
uadmin@ubuntu1:~/devops3/flask_mysql_app$ docker build -t pritiimaurya/devops-assignment3:v2 .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
Install the buildx component to build images with BuildKit:
https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  14.34kB
Step 1/12 : FROM ubuntu
--> 53958ec7b67c
```

6. Pushing the new docker image

```
Successfully tagged pritimaurya/devops-assignment3:v2
uadmin@ubuntu1:~/devops3/flask_mysql_app$ docker push pritimaurya/devops-assignment3:v2
The push refers to repository [docker.io/pritimaurya/devops-assignment3]
96729d8d5401: Layer already exists
62e215686e33: Pushed
d1f56e4c7f2f: Layer already exists
7ade7b7856ac: Layer already exists
1bd0c9faeb55: Layer already exists
3e49383002cf: Layer already exists
81e2f2053c8f: Layer already exists
331d1419ab3e: Layer already exists
7873277a23e7: Layer already exists
1f65d64983f1: Layer already exists
d19c9e709049: Layer already exists
v2: digest: sha256:328fc58c6f2e12b6eed2039c113139a999055386d2620f805a013bc4dfc99be2 size: 2917
```

7. setting the new image for the deployment

```
uadmin@ubuntu1:~/devops3/flask_mysql_app$ kubectl set image deployment/flask-app
devops-assignment3=pritimaurya/devops-assignment3:v2
deployment.apps/flask-app image updated
uadmin@ubuntu1:~/devops3/flask_mysql_app$ kubectl rollout status deployment/flask
-app
deployment "flask-app" successfully rolled out
uadmin@ubuntu1:~/devops3/flask_mysql_app$
```

8. Updated website

